

IPC USING SHARED MEMORY:

(I) Sender.c

```
#include <sys/types.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#define SharedMemSize 50
int main()
{
    int shmid;
    key_t key;
    char *shared_memory;
    key = 5677;
    // Create shared memory segment
    if ((shmid = shmget(key, SharedMemSize, IPC_CREAT | 0666)) < 0)
    {
        perror("shmget");
        exit(1);
    }
    // Attach the segment
    if ((shared_memory = shmat(shmid, NULL, 0)) == (char *) -1)
    {
        perror("shmat");
        exit(1);
    }
    // Write message to shared memory
    sprintf(shared_memory, "Welcome to Shared Memory");
    // Give time for receiver to read
    sleep(2);

    return 0;
}
```

(II) *Receiver.c*

```
#include <sys/types.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <stdio.h>
#include <stdlib.h>
#define SharedMemSize 5
int main()
{
    int shmid;
    key_t key;
    char *shared_memory;
    key = 5677;
    // Get shared memory segment
    if ((shmid = shmget(key, SharedMemSize, 0666)) < 0)
    {
        perror("shmget");
        exit(1);
    }
    // Attach the segment
    if ((shared_memory = shmat(shmid, NULL, 0)) == (char *) -1)
    {
        perror("shmat");
        exit(1);
    }
    // Read message from shared memory
    printf("Message Received: %s\n", shared_memory);
    return 0;
}
```